1.0 Vectors

1.1 Scalar quantities

Quantities which can be entirely determined by numbers with appropriate units are called scalar quantities.
Distance, time, mass, volume, temperature are scalar quantities.
Further, two quantities of the same kind, when added will give another quantity of the same kind.

Examples:

Mass is 10 kg; Temperature is 27° C, Time is 20 s. Length is 2 m; Area is 5 m^2, Volume is 4 m^3, capacity is 2 *l*, speed is 5 m s^{-1}. The numerical parts of the above examples without the units are called **Scalars.**

There are also quantities which cannot be described fully by magnitude (with units) alone but which can be known fully by magnitute and direction. For example,

i) A ship is travelling with a speed of 15 km h^{-1} due north.
ii) A force 20 newton act on a particle vertically downwards.

Study on vectors was first focused in middle of 19th century. In the recent past "Vectors" has become an indispensable tool, used in the mathematical calculation of engineers, mathematician and physicists while physical and geometrical problems can be expressed concisely, by using vectors.

1.2 Vector quantities

Quantities which can be described completly by magnitudes (with units) and directions are called vector quantitics.

Examples:

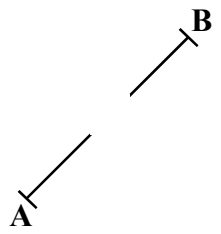
i. Displacement due north is 5 m.
ii. Velocity is 15 m s^{-1} due south east.
iii. Weight is 30 N, Vertically downwards.
iv. Force of 10 N inclined upwards 30° to the horizontal.
Vectors have both magnitude and direction.

1.3 Representation of vectors

There are two ways of representing vectors.

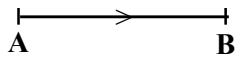
Geometrical Representation

A vector can be represented by a **directed line segment** $\overrightarrow{AB}$
The length of the line segment will give the magnitude of the vector and the arrow head on it denotes the direction. This is said to be the geometrical representation of a vector.



Example:

To denote a force of 4 N due east a straight line segment AB is drawn towards east where AB = 4 units. The direction of the force is denoted by the arrow from A to B as shown below.



Algebraic Representation

The vector $\overrightarrow{AB}$ is denoted by a single algebraic symbol such as $\underline{a}$ or $\bar{a}$. In some text books generally it is denoted by the symbol **a** in dark print.

1.4 Modulus of a vector

The magnitude of a vector is known as its modulus.

The modulus of a vector $\overrightarrow{AB}$ **(**or $\underline{a}$**)** is denote by $\left|\overrightarrow{AB}\right|$ or $|\underline{a}|$

The modulus of a vector is always non- negative.

1.5 Equality of two vectors

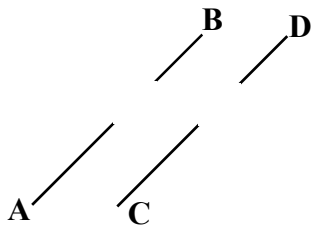
If two vectors are equal in magnitude and are in the same direction they are called equal vectors.

The two vectors $\overrightarrow{AB}$ **(** $= \underline{a}$**)** and $\overrightarrow{CD}$ $(=\underline{b})$ are equal if and only if

i) $\left|\overrightarrow{AB}\right| = \left|\overrightarrow{CD}\right|$

ii) AB // CD and

(iii) $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are in the same direction.



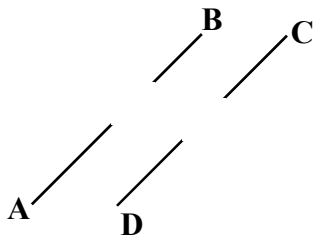
Note : Consider the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$

AB = CD i.e $\left|\overrightarrow{AB}\right| = \left|\overrightarrow{CD}\right|$

AB // CD

But they are not in the same direction.

Therefore $\overrightarrow{AB} \neq \overrightarrow{CD}$; $\underline{a} \neq \underline{b}$



1.6 Unit vector

A vector with unit magnitude is called unit vector. Given a vector $\underline{a}$, the unit vector in the direction of $\boldsymbol{a}$ is $\frac{\underline{a}}{|\underline{a}|}$ is denoted by $\hat{\underline{a}}$.

1.7 Zero vector (null vector)

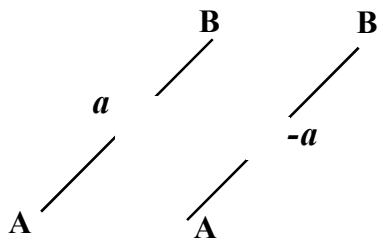
A vector with zero magnitude is called zero vector. It is denoted by **0**. $|\mathbf{0}| = 0$ and its direction is arbitrary and is represented by a point.

1.8 Negative vector of a given vector

Given a vector $\overrightarrow{AB}$, the vector $\overrightarrow{BA}$ is negative vector of $\overrightarrow{AB}$ and is written $\overrightarrow{BA} = -\overrightarrow{AB}$.

If $\overrightarrow{AB} = \underline{a}$, then $\overrightarrow{BA} = -\underline{a}$

$|\overrightarrow{AB}| = |\overrightarrow{BA}|$, $|\underline{a}| = |-\underline{a}|$



1.9 Scalar multiple of a vector

When $\underline{a}$ is a vector and λ is a scalar, then $\lambda\underline{a}$ is the product of the vector $\underline{a}$ and scalar λ. Here λ should be considerd under three cases namely when $\lambda > 0$, $\lambda = 0$ and $\lambda < 0$.

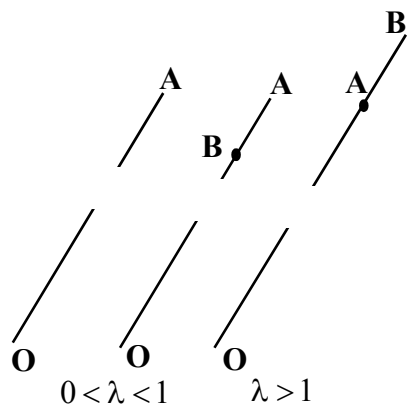
Case (i) $\lambda > 0$

Let $\overrightarrow{OA} = \underline{a}$,

Take a point B on OA.

(or produced OA) such that OB = λOA

$\overrightarrow{OB} = \lambda\overrightarrow{OA} = \lambda\underline{a}$

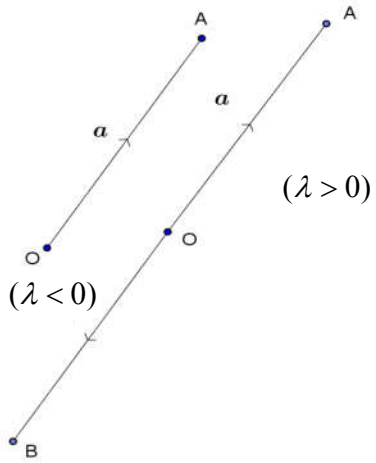


(ii) When $\lambda = \mathbf{0}$, $\lambda\underline{a}$ is defind as the nullvector.

That is $\lambda\underline{a} = 0\underline{a} = \mathbf{0}$

(iii) $\lambda < 0$

In this case $\lambda\underline{a}$ is a vector opposite to the direction of **a** with a magnitude of $|\lambda|$ times OA. Choose a point B on AO produced such that OB = $|\lambda|$ OA. Then $\overrightarrow{OB} = \lambda\underline{a}$.

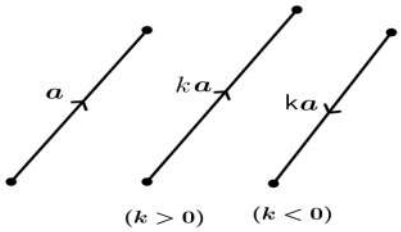


1.10 Parallel vectors

Given a vector $\underline{a}$ and k$\underline{a}$, k$\underline{a}$ is a vector parallel to $\underline{a}$

(i) When k > **0**, the vector k$\underline{a}$ is in the direction of $\underline{a}$

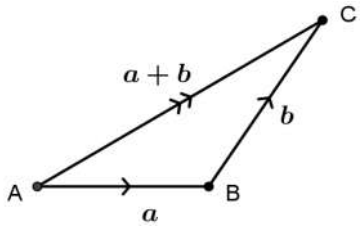
(ii) When k < **0**, the vector k$\underline{a}$ is opposite in direction to that of $\underline{a}$.



Two vectors ***a*** and ***b*** are said to be parallel. if $\underline{b} = \lambda\underline{a}$

1.11 Vector addition

If two vectors $\underline{a}$ and $\underline{b}$ are represented by $\overrightarrow{AB}$ and $\overrightarrow{BC}$ respectively then the vector addtion of $\underline{a}$ and $\underline{b}$ is represented by $\overrightarrow{AC}$



ie. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$

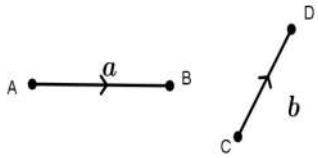
$= \underline{a} + \underline{b}$

This is called the triangle law of vector addition.

Let $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{CD} = \underline{b}$ be two vectors.

Draw a line segment PQ such that PQ = AB and PQ // AB.

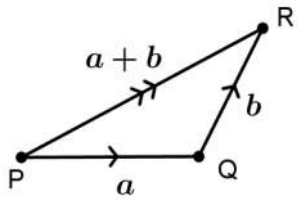
Draw a line segment QR such that QR = CD and QR // CD.



By definition $\overrightarrow{AB} = \overrightarrow{PQ} = \underline{a}$ and $\overrightarrow{CD} = \overrightarrow{QR} = \underline{b}$

According to the triangle of law of addition

$\overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{QR} = \underline{a} + \underline{b}$



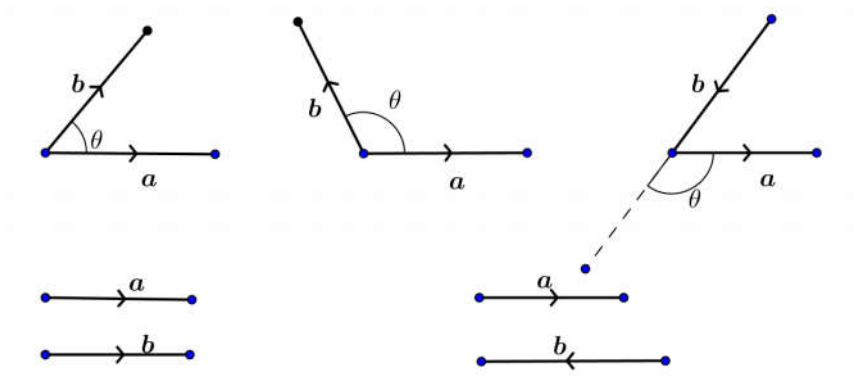
1.12 Definition of a vector

A vector has magnitude and direction and obeys the triangle of law of addition.

1.13 Angle between two vectors

Let $\underline{a}$ and $\underline{b}$ be two vectors.

The angle θ between $\underline{a}$ and $\underline{b}$ is shown below.



Note that $0 \le \theta \le \pi$

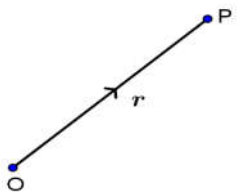
If $\underline{a}$ and $\underline{b}$ are parallel and are in the same direction, then $\theta = 0$.

If $\underline{a}$ and $\underline{b}$ are parallel and are in the opposite direction, then $\theta = \pi$.

1.14 Position vector

With a fixed point O chosen as the origin, the position of any point P can be denoted by the vector $\overrightarrow{OP}$.

The vector $\overrightarrow{OP} = \underline{r}$ is (known as the) position vector of P with respect to O.



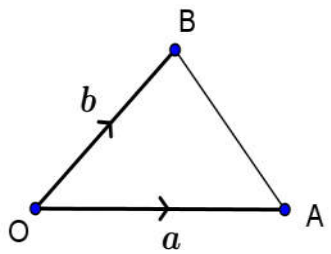
Let the position vectors of two points A and B be $\underline{a}$ and $\underline{b}$.

$\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$

$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

$= \underline{b} - \underline{a}$



1.15 Laws of vector algebra

Let $\underline{a}$, $\underline{b}$, $\underline{c}$ be vectors and λ, μ be scalars.

(i) $\underline{a} + \underline{b} = \underline{b} + \underline{a}$ (Commutative Law)

(ii) $(\underline{a} + \underline{b}) + \underline{c} = \underline{a} + (\underline{b} + \underline{c})$ (Associative Law)

(iii) $\lambda(\underline{a} + \underline{b}) = \lambda\underline{a} + \lambda\underline{b}$ (Distributive Law)

(iv) $\underline{a} + \underline{0} = \underline{a} = \underline{0} + \underline{a}$

(v) $\underline{a} + (-\underline{a}) = \underline{0} = (-\underline{a}) + \underline{a}$

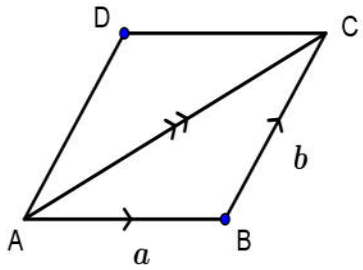
(vi) $(\lambda + \mu)\underline{a} = \lambda\underline{a} + \mu\underline{a}$

(vii) $\lambda\mu(\underline{a}) = \lambda[\mu\underline{a}] = \mu[\lambda\underline{a}]$

proof :

(i) Let $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{BC} = \underline{b}$

Complete the parallelogram ABCD



Now $\overrightarrow{DC} = \overrightarrow{AB} = \underline{a}$

$\overrightarrow{AD} = \overrightarrow{BC} = \underline{b}$

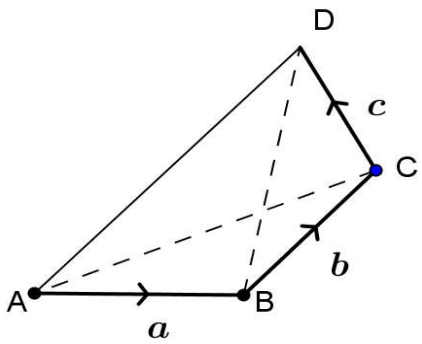
By triangle law of vector addition

$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \underline{a} + \underline{b}$

$\overrightarrow{AC} = \overrightarrow{AD} + \overrightarrow{DC} = \underline{b} + \underline{a}$

Hence $\underline{a} + \underline{b} = \underline{b} + \underline{a}$

(ii) Let $\overrightarrow{AB} = \underline{a}$, $\overrightarrow{BC} = \underline{b}$ and $\overrightarrow{CD} = \underline{c}$



$\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BD}$

$= \overrightarrow{AB} + (\overrightarrow{BC} + \overrightarrow{CD})$

$= \underline{a} + (\underline{b} + \underline{c})$ ..... ①

$\overrightarrow{AD} = \overrightarrow{AC} + \overrightarrow{CD}$

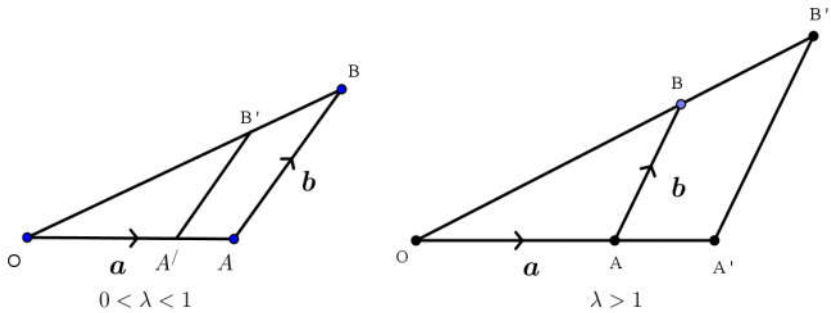
$= (\overrightarrow{AB} + \overrightarrow{BC}) + \overrightarrow{CD}$

$= (\underline{a} + \underline{b}) + \underline{c}$ .....②

From (1) and (2) $(\underline{a} + \underline{b}) + \underline{c} = \underline{a} + (\underline{b} + \underline{c})$

(iii) $\lambda(\underline{a} + \underline{b}) = \lambda\underline{a} + \lambda\underline{b}$

Let $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{AB} = \underline{b}$



Take the point A' on OA (or OA produced)

such that $\overrightarrow{OA'} = \lambda\overrightarrow{OA} = \lambda\underline{a}$

The line drawn parallel to AB through A' meet OB (or OB produed) at B'.

Now, ΔOAB, $\Delta OA'B'$ are similar triangles.

$$\frac{OA'}{OA} = \frac{A'B'}{AB} = \frac{OB'}{OB} = \lambda.$$

$\overrightarrow{A'B'} = \lambda\overrightarrow{AB} = \lambda\underline{b}$ and $\overrightarrow{OB'} = \lambda\overrightarrow{OB}$ ..... ①

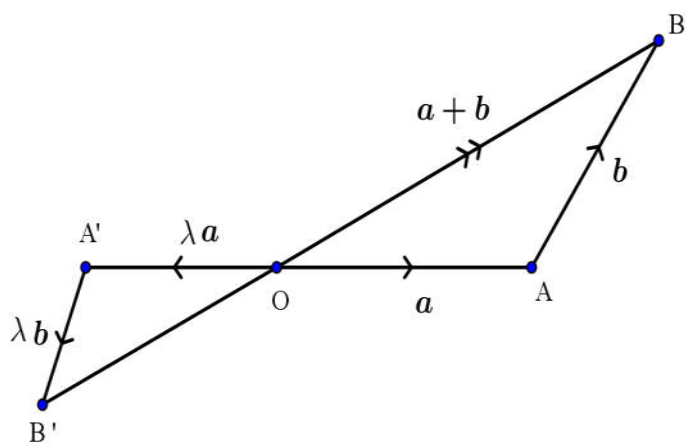
$\overrightarrow{OB'} = \overrightarrow{OA'} + \overrightarrow{A'B'} = \lambda\underline{a} + \lambda\underline{b}$ .....②

$\lambda\overrightarrow{OB} = \lambda(\overrightarrow{OA} + \overrightarrow{AB}) = \lambda(\underline{a} + \underline{b})$ .....③

From (i) $\overrightarrow{OB'} = \lambda\overrightarrow{OB}$

$\lambda\underline{a} + \lambda\underline{b} = \lambda(\underline{a} + \underline{b})$

When $\lambda < 0$



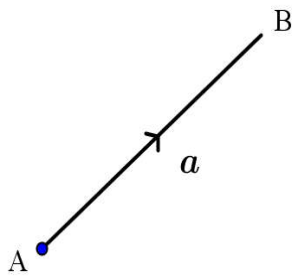
$\overrightarrow{OA} = \underline{a}$, $\overrightarrow{AB} = \underline{b}$, $\overrightarrow{OA'} = \lambda\underline{a}$

A'B' is drawn parallel to BA and meets BO produced at B'. $\overrightarrow{A'B'} = \lambda\underline{b}$ and $\overrightarrow{OB'} = \lambda\underline{a} + \lambda\underline{b}$

By the properties of similar triangles and vectors it can be easily proved that

$\lambda(\underline{a} + \underline{b}) = \lambda\underline{a} + \lambda\underline{b}$ when $\lambda < 0$

(iv) $\underline{a} + \underline{0} = \underline{a} = \underline{0} + a$



Let $\overrightarrow{AB} = \underline{a}$

$\overrightarrow{AB} = \overrightarrow{AB} + \overrightarrow{BB}$

$\underline{a} = \underline{a} + \underline{0}$ ..... ①

$\overrightarrow{AB} = \overrightarrow{AA} + \overrightarrow{AB}$

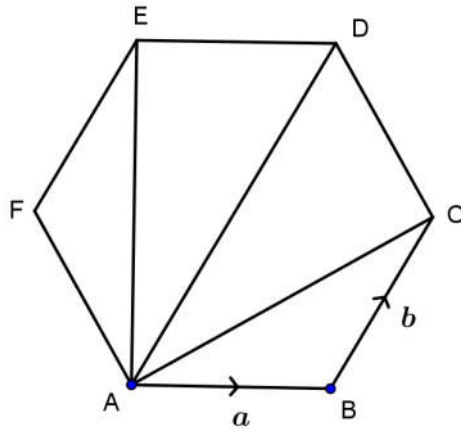
$\underline{a} = \underline{0} + \underline{a}$ ..... ②

From (1) and (2) $\underline{a} + \underline{0} = \underline{a} = \underline{0} + \underline{a}$

1.16 Worked examples

Example 1

ABCDEF is a regular hexagon. If $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{BC} = \underline{b}$, express the vectors $\overrightarrow{AC}$, $\overrightarrow{AD}$, $\overrightarrow{AE}$ and $\overrightarrow{AF}$ in terms of $\underline{a}$, $\underline{b}$



$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \underline{a} + \underline{b}$ .....①

By geometry AD = 2BC; AD // BC

$\therefore\ \overrightarrow{AD} = 2\overrightarrow{BC} = 2\underline{b}$ .....②

$\overrightarrow{AE} = \overrightarrow{AD} + \overrightarrow{DE}$

$= 2\underline{b} + (-\underline{a}) = 2\underline{b} - \underline{a}$ .....③

By geometry BC = FE; BC // FE

$\overrightarrow{FE} = \overrightarrow{BC} = \underline{b}$

$\overrightarrow{AF} = \overrightarrow{AE} + \overrightarrow{EF} = (2\underline{b} - \underline{a}) - \underline{b} = \underline{b} - \underline{a}$

Example 2

The position vectors of A and B are $\underline{a}$ and $\underline{b}$ respectively

(i) C is the midpoint of AB.

(ii) D is a point on AB such that AD : DB = 1 : 2

(iii) E is a point on AB such that AE : EB = 2 : 1

Find the position vectors of C, D and E

Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$. Then $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \underline{b} - \underline{a}$

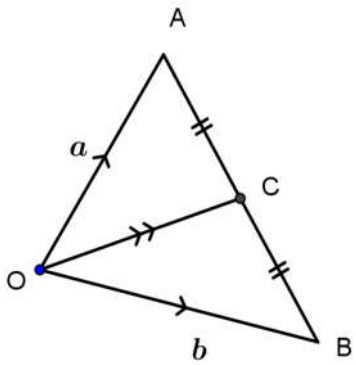
(i) AC = CB

$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC}$

$= \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB}$

$= \underline{a} + \frac{1}{2}(\underline{b} - \underline{a})$

$= \frac{1}{2}(\underline{a} + \underline{b})$



(ii) AD : DB = 1 : 2

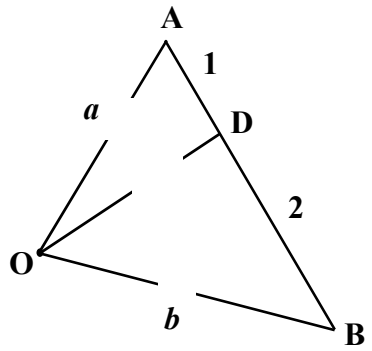
$\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD}$

$= \overrightarrow{OA} + \frac{1}{3}\overrightarrow{AB}$

$= \overrightarrow{OA} + \frac{1}{3}\left(\overrightarrow{OB} - \overrightarrow{OA}\right)$

$= \underline{a} + \frac{1}{3}(\underline{b} - \underline{a})$

$= \frac{2}{3}\underline{a} + \frac{1}{3}\underline{b} = \frac{1}{3}(2\underline{a} + \underline{b})$



(iii) $\overrightarrow{OE} = \overrightarrow{OA} + \overrightarrow{AE}$

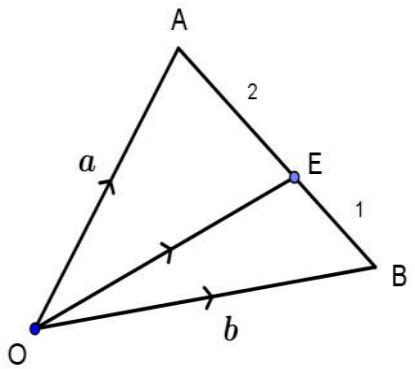
$= \overrightarrow{OA} + \frac{2}{3}\overrightarrow{AB}$

$= \overrightarrow{OA} + \frac{2}{3}\left(\overrightarrow{OB} - \overrightarrow{OA}\right)$

$= \underline{a} + \frac{2}{3}(\underline{b} - \underline{a})$

$= \frac{1}{3}\underline{a} + \frac{2}{3}\underline{b}$

$= \frac{1}{3}(\underline{a} + 2\underline{b})$



Example 3

Let $-2\underline{p} + 5\underline{q}$, $7\underline{p} - \underline{q}$ and $\underline{p} + 3\underline{q}$ be the position vectors of three points A, B and C respectively, with respect to a fixed origin O, where $\underline{p}$ and $\underline{q}$ are two non-parallel vectors. Show that the points A, B and C are collinear and find the ratio in which C divides AB.

$\overrightarrow{OA} = -2\underline{p} + 5\underline{q}$, $\overrightarrow{OB} = 7\underline{p} - \underline{q}$, $\overrightarrow{OC} = \underline{p} + 3\underline{q}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

$= (7\underline{p} - \underline{q}) - (-2\underline{p} + 5\underline{q})$

$= 9\underline{p} - 6\underline{q}$

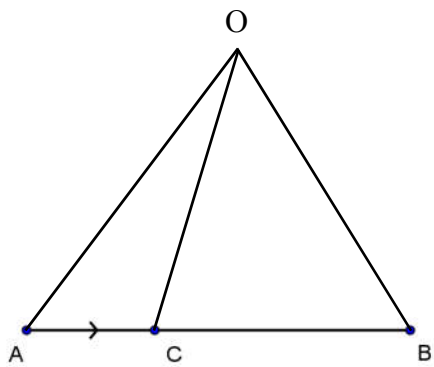
$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$

$= (\underline{p} + 3\underline{q}) - (-2\underline{p} + 5\underline{q})$

$= 3\underline{p} - 2\underline{q}$

$\overrightarrow{AB} = 3(3\underline{p} - 2\underline{q})$

$\overrightarrow{AC} = 3\underline{p} - 2\underline{q} \quad \Rightarrow \quad = \overrightarrow{AB} = 3\overrightarrow{AC}$



Therefore A, B and C are collinear and AC : CB = 1 : 2

Example 4

$\underline{a}$, $\underline{b}$ are two non-zero and non parallel vectors and α, β are scalars. Prove that $\alpha\underline{a} + \beta\underline{b} = \underline{0}$ if and only if $\alpha = 0$ and $\beta = 0$.

Assume that $\alpha = 0$ and $\beta = 0$.

$\alpha\underline{a} + \beta\underline{b} = \underline{0} + \underline{0} = \underline{0}$.

Conversely let $\alpha\underline{a} + \beta\underline{b} = \underline{0}$

case (i) : Suppose that $\alpha = 0$

Then $\underline{0} + \beta\underline{b} = \underline{0}$

$\beta\underline{b} = \underline{0}$

since $\underline{b} \neq \underline{0}$, it follows that $\beta = 0$

If $\alpha = 0$, then $\beta = 0$

Similarly we can show that if $\beta = 0$, then $\alpha = 0$

case (ii) : Suppose that $\alpha \neq 0$

$\alpha\underline{a} + \beta\underline{b} = \underline{0}$

$\alpha\underline{a} = -\beta\underline{b}$

$\underline{a} = -\dfrac{\beta}{\alpha}\underline{b}$ $\qquad (\alpha \neq \underline{0})$

The above equation implies that $\underline{a} \,//\, \underline{b}$

This is a contradiction

Hence $\alpha = 0$ and from the first part $\beta = 0$

ie. $\alpha\underline{a} + \beta\underline{b} = \underline{0}$ if and only if $\alpha = 0$, $\beta = 0$

Example 5

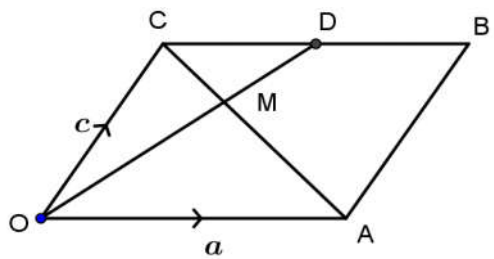
OABC is a parallelogram. D is the midpoint of BC. OD and AC intersect at M. Given that $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OC} = \underline{c}$

(i) Find $\overrightarrow{OD}$ interms of $\underline{a}$ and $\underline{c}$

(ii) If OM : MD = λ : 1, find $\overrightarrow{OM}$ in terms of $\underline{a}$, $\underline{c}$ and λ

(iii) If AM : MC = μ : 1, find $\overrightarrow{AM}$ in terms of $\underline{a}$, $\underline{c}$ and μ and hence find $\overrightarrow{OM}$

(iv) Using the results obtained in (ii) and (iii) above find the values of λ and μ.



(i) $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OC} = \underline{c}$; $\overrightarrow{OA} = \overrightarrow{CB} = \underline{a}$

$$\overrightarrow{OD} = \overrightarrow{OC} + \overrightarrow{CD}$$
$$= \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CB}$$
$$= \underline{c} + \frac{1}{2}\underline{a} \quad \text{.....} ①$$

OM : MD = λ : 1, $\overrightarrow{OM} = \frac{\lambda}{\lambda+1}\overrightarrow{OD} = \frac{\lambda}{\lambda+1}\left[\frac{1}{2}\underline{a}+\underline{c}\right]$ ..... ②

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$$
$$= \underline{c} - \underline{a}$$

AM : MC = μ : 1, $\overrightarrow{AM} = \frac{\mu}{\mu+1}\overrightarrow{AC} = \frac{\mu}{\mu+1}(\underline{c} - \underline{a})$

$$\overrightarrow{OM} = \overrightarrow{OA} + \overrightarrow{AM} = \underline{a} + \frac{\mu}{\mu+1}(\underline{c} - \underline{a})$$
$$= \left(1-\frac{\mu}{\mu+1}\right)\underline{a} + \frac{\mu}{\mu+1}\underline{c}$$
$$= \frac{1}{\mu+1}\underline{a} + \frac{\mu}{\mu+1}\underline{c} \quad \text{.....} ③$$

From ② and ③

Since $\underline{a}$ is not parallel to $\underline{c}$

$$\frac{\lambda}{2(\lambda+1)} = \frac{1}{\mu+1} \quad \text{.....} ④$$

$$\frac{\lambda}{\lambda+1} = \frac{\mu}{\mu+1} \quad \text{.....} ⑤$$

$\frac{④}{⑤}$ gives $\frac{1}{2} = \frac{1}{\mu}$, μ = 2

If μ = 2, from ④ $\frac{\lambda}{2(\lambda+1)} = \frac{1}{3}$

λ = 2

λ = 2 = μ

ie. OM : MD = AM : MC = 2 : 1

1.17 Exercises

1. ABCDEF is a reqular hexagon. $\overrightarrow{AB} = \underline{a}$, $\overrightarrow{AC} = \underline{b}$ Find $\overrightarrow{AD}$, $\overrightarrow{AE}$, $\overrightarrow{AF}$ in terms of $\underline{a}$, $\underline{b}$.

2. ABCDEF is a regular hexagon and O is its centre. If $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ find $\overrightarrow{AB}$, $\overrightarrow{BC}$, $\overrightarrow{CD}$, $\overrightarrow{DE}$, $\overrightarrow{EF}$, $\overrightarrow{FA}$ in terms of $\underline{a}$, $\underline{b}$

3. ABCD is a plane quadrilatelral and O is apoint in the plane of the quadrilateral. If $\overrightarrow{AO} + \overrightarrow{CO} = \overrightarrow{DO} + \overrightarrow{BO}$, show that ABCD is a parallelogram.

4. ABC is an isosceles triangle with BA = BC and D is the midpoint of AC. Show that $\overrightarrow{BA} + \overrightarrow{BC} = 2\overrightarrow{BD}$.

5. $\underline{a}$ and $\underline{b}$ are two vectors perpendicular to each other. Using triangle of law of vector addition show that $|\underline{a}+\underline{b}| = |\underline{a}-\underline{b}|$. When $|\underline{a}-\underline{b}| = 5$ and $|\underline{a}| = 3$, find $|\underline{b}|$.

6. $\underline{a}$, $\underline{b}$ are two vectors. such that $|\underline{a}| = 6$, $|\underline{b}| = 6$ and the angle between $\underline{a}$ and $\underline{b}$ is 60°. Find $|\underline{a}+\underline{b}|$ and $|\underline{a}-\underline{b}|$.

Use vectors to prove the following questions (7,8,9).

7. ABC is a triangle. D and E are the midpoints of AB and AC. Prove that DE = $\frac{1}{2}$ BC and DE is parallel to BC.

8. ABCD is a quadrilateral. P, Q, R and S are the midpoints of AB, BC, CD and DA respectively. Show that PQRS is a parallelogram.

9. ABC is a triangle. The position vectors of A, B and C are $\underline{a}$, $\underline{b}$, $\underline{c}$ respectively. Find the position vector of the centroid of the triangle ABC.

10. OABC is a parallelogram. D is the midpoint of AB. OD and AC intersects at E. $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, OE : ED = λ : 1, CE : EA = μ : 1.

    i. Find $\overrightarrow{OD}$ in terms of $\underline{a}$ and $\underline{b}$. Hence write the vector $\overrightarrow{OE}$ in terms of λ, $\underline{a}$ and $\underline{b}$.

    ii. Find the vector $\overrightarrow{AC}$ and write the vector $\overrightarrow{OE}$ in terms of μ, $\underline{a}$ and $\underline{b}$.

    iii. Using the results obtained in (i) and (ii) above find λ and μ.

    iv. When OD and CB produced meet at H, find $\overrightarrow{OH}$.

11. Let OABC be a quadrilateral and let D and E be the midpoints of the diagonal OB and AC respectively. Also let F be the mid-point of DE. By taking the position vectors of the points A, B and C with respect to O be $\underline{a}$, $\underline{b}$ and $\underline{c}$ respectively, show that $\overrightarrow{OF} = \frac{1}{4}(\underline{a}+\underline{b}+\underline{c})$.

    Let P and Q be the midpoints of the sides OA and BC respectively. Show that the points. Show that the points P, F and Q are collinear and find the ratio PF : FQ.

12. Let A and B two distinct points not collinear with a point O. The position vectors of A and B with respect to the point O is $\underline{a}$ and $\underline{b}$. If D is the point on AB such that BD = 2DA. Show that the position vector of D with respect to point O is $\frac{1}{3}(2\underline{a}+\underline{b})$.

If $\overrightarrow{BC} = K\underline{a}$ (K > 1) and the points O, D and C are collinear, find the value of **K** and the ratio OD : DC Express $\overrightarrow{AC}$ in terms of $\underline{a}$ and $\underline{b}$.
Further if the line through O parallel to AC meets AB at E show that 6DE = AB.

13. Let ABCD is a trapezium such that $\overrightarrow{DC} = \frac{1}{2}\overrightarrow{AB}$. Also let $\overrightarrow{AB} = \underline{p}$ and $\overrightarrow{AD} = \underline{q}$. The point E lies on BC such that $\overrightarrow{BE} = \frac{1}{3}\overrightarrow{BC}$. The point of intersection F of AE and BD satisfies $\overrightarrow{BF} = \lambda\overrightarrow{BD}$. where λ $(0 < \lambda < 1)$ is a constant. Show that $\overrightarrow{AF} = (1-\lambda)\underline{p}+\lambda\underline{q}$. Hence find the value of λ.

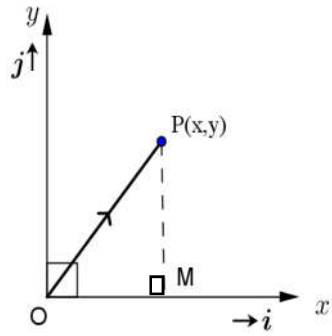
1.18 Cartesian vector notation

Consider the cartesian plane ***xoy***.
Let the unit vector in the direction O*x* be ***i***, the unit vector in the direction O*y* be ***j***, and P $\equiv (x, y)$

Let $\overrightarrow{OP} = \underline{r}$

$\underline{r} = \overrightarrow{OP} = \overrightarrow{OM}+\overrightarrow{MP} = x\underline{i} + y\underline{j}$ $\quad (|\overrightarrow{OM}| = OM = x,\ |\overrightarrow{MP}| = MP = y)$

$|\underline{r}| = OP = \sqrt{x^2+y^2}$



Let $\underline{a} = a_1\underline{i} + a_2\underline{j}$ and $\underline{b} = b_1\underline{i} + b_2\underline{j}$

$$\underline{a}+\underline{b} = (a_1\underline{i}+b_1\underline{i})+(a_2\underline{j}+b_2\underline{j})$$
$$= (a_1+b_1)\underline{i}+(a_2+b_2)\underline{j}$$

Proof

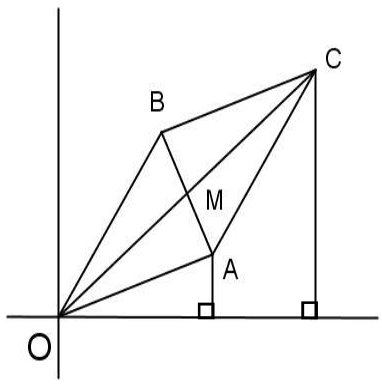
$\overrightarrow{OA} = a_1\underline{i}+a_2\underline{j}$ $\qquad A \equiv (a_1, a_2)$

$\overrightarrow{OB} = b_1\underline{i}+b_2\underline{j}$ $\qquad B \equiv (b_1, b_2)$

OACB is a parallelogram $\overrightarrow{OC} = \overrightarrow{OA}+\overrightarrow{AC} = \underline{a}+\underline{b}$

Since M is the midpoint of AB, $M \equiv \left(\frac{a_1+b_1}{2}, \frac{a_2+b_2}{2}\right)$

M is the midpoint of OC. $\quad C \equiv (a_1+b_1, a_2+b_2)$



$$\overrightarrow{OC} = (a_1+b_1)\underline{i}+(a_2+b_2)\underline{j}$$

If $\underline{a} = a_1 i + a_2 j$ and $\underline{b} = b_1 i + b_2 j$ then

$\underline{a} - \underline{b} = a + (-b) = (a_1\underline{i} + a_2\underline{j}) + (-b_1\underline{i} - b_2\underline{j})$

$\underline{a} - \underline{b} = a + (-b) = (a_1\underline{i} + a_2\underline{j}) + (-b_1\underline{i} - b_2\underline{j})$

$= (a_1 - b_1)\underline{i} + (a_2 - b_2)\underline{j}$

Example 6

If A ≡ (2, -1) and B ≡ (5, 3) find

i. $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{AB}$ in terms of $\underline{i}, \underline{j}$

ii. $\left|\overrightarrow{OA}\right|$, $\left|\overrightarrow{OB}\right|$, $\left|\overrightarrow{AB}\right|$

iii. the unit vector in the direction $\overrightarrow{AB}$

A ≡ (2, -1), B ≡ (5, 3)

(i) $\overrightarrow{OA} = 2\underline{i} - \underline{j}$ $\qquad \overrightarrow{OB} = 5\underline{i} + 3\underline{j}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (5\underline{i} + 3\underline{j}) - (2\underline{i} - \underline{j})$

$= 3\underline{i} + 4\underline{j}$

(ii) $\overrightarrow{OA} = 2\underline{i} - \underline{j}$, $\qquad \left|\overrightarrow{OA}\right| = \sqrt{2^2+1^2} = \sqrt{5}$

$\overrightarrow{OB} = 5\underline{i} + 3\underline{j}$, $\qquad \left|\overrightarrow{OB}\right| = \sqrt{5^2+3^2} = \sqrt{34}$

$\overrightarrow{AB} = 3\underline{i} + 4\underline{j}$, $\qquad \left|\overrightarrow{AB}\right| = \sqrt{3^2+4^2} = \sqrt{25} = 5$

(iii) unit vector in the direction $\overrightarrow{AB}$ is

$$\frac{\overrightarrow{AB}}{\left|\overrightarrow{AB}\right|} = \frac{1}{5}(3\underline{i} + 4\underline{j})$$

1.19 Exercises

1. Let $\underline{a} = \underline{i} - 2\underline{j}$, $\quad b = 4\underline{j}$, $\quad \underline{c} = 3\underline{i} - \underline{j}$ $\quad$ Find

   i. (a) $2\underline{a} + \underline{b}$ (b) $\underline{a} + 3\underline{c}$ (c) $2\underline{a} - \underline{b} - \underline{c}$

   ii. (a) $|2\underline{a}+\underline{b}|$ (b) $|\underline{a}+3\underline{c}|$ (c) $|2\underline{a}-\underline{b}-\underline{c}|$

   iii. the unit vector in the direction $\underline{a}+\underline{b}+\underline{c}$

2. Given that A ≡ (4, 3), B ≡ (6, 6) and C ≡ (0, 1)

   (a) Write the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$

   (b) Find $\overrightarrow{AB}$, $\overrightarrow{BC}$, $\overrightarrow{CA}$

   (c) Find $\left|\overrightarrow{AB}\right|$, $\left|\overrightarrow{BC}\right|$, $\left|\overrightarrow{CA}\right|$

3. O is the origin and $\overrightarrow{OA} = -\underline{i} + 5\underline{j}$, $\quad \overrightarrow{OB} = 2\underline{i} + 4\underline{j}$ and $\overrightarrow{OC} = 2\underline{j}$. Find $\overrightarrow{AB}$, $\overrightarrow{BC}$, $\overrightarrow{CA}$. Hence show that ABC is an isosceles triangle.

4. If $\overrightarrow{OA} = \underline{i} + 2\underline{j}$, $\overrightarrow{OB} = 3\underline{i} - \underline{j}$ and $\overrightarrow{OC} = -\underline{i} + 5\underline{j}$. Find $\overrightarrow{AB}$ and $\overrightarrow{CA}$ and hence show that the points A, B and C are collinear.

5. The position vectors of the points A and B are $\underline{a}$ and $\underline{b}$ respectively, where $\underline{a} = 2\underline{i} + 3\underline{j}$ *and* $\underline{b} = \underline{i} + 5\underline{j}$.
   (i) If R is the midpoint of AB, find the position vector of R interms of $\underline{i}, \underline{j}$.
   (ii) IF $\underline{c} = 2\underline{a} - \underline{b}$. Find the unit vector along $\underline{c}$ interms of $\underline{i}, \underline{j}$

6. (a) Find in the form $a\underline{i} + b\underline{j}$, a vector of magnitude 10 units in the direction $3\underline{i} - 4\underline{j}$
   (b) A ≡ (-2, -5) and B ≡ (3, 7)
      (i) Write $\overrightarrow{OA}$, $\overrightarrow{OB}$ and hence $\overrightarrow{AB}$
      (ii) Find in the form $a\underline{i} + b\underline{j}$, a vector of magnitude 65 units in the diretion $\overrightarrow{AB}$

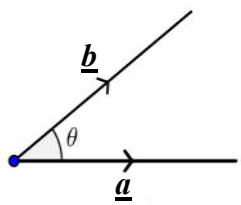
1.20 Scalar product of two vectors

We learnt earlier the rules of vector addition and subtraction. Two types of products have been defined.
(i) The scalar product of two vectors.
(ii) The vector product of two vectors.

The scalar product is also known as the dot product. The result of a dot products a scalar and the result of the vector product is a vector.

Definition : **Scalar Product**

Let $\underline{a}$ and $\underline{b}$ be any two non-zero vectors and θ be the angle between the two vectors. The scalar product of two vectors $\underline{a}$ and $\underline{b}$ is dfined as $\underline{a} \cdot \underline{b} = |\underline{a}|\ |\underline{b}| \cos\theta$ ($0 \le \theta \le \pi$)



Properties of the scalar product

1. $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{a}$ (Commutative Law)

By definition, $\underline{a} - \underline{b} = |\underline{a}|\ |\underline{b}| \cos\theta$

$= |\underline{b}|\ |\underline{a}| \cos\theta$

Hence $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{a}$

2. If $\underline{a}$ and $\underline{b}$ are two non - zero vectors $\underline{a} \cdot \underline{b} = 0$. if and only if $\underline{a}$ is perpenticular to $\underline{b}$.

$$\underline{a} \cdot \underline{b} = 0 \Leftrightarrow |\underline{a}|\ |\underline{b}| \cos\theta = 0$$

$$\Leftrightarrow \cos\theta = 0 \qquad (\underline{a}, \underline{b} \neq \underline{0})$$

$$\Leftrightarrow \theta = \frac{\pi}{2}$$

3. $\underline{a} \cdot \underline{a} = |\underline{a}|\ |\underline{a}| \cos 0 = |\underline{a}|^2$ also writen $\underline{a}^2$

$\underline{i} \cdot \underline{i} = |\underline{i}|\ |\underline{i}| \cos 0 = 1 \times 1 \times 1 = 1$

$\underline{j} \cdot \underline{j} = |\underline{j}|\ |\underline{j}| \cos 0 = 1 \times 1 \times 1 = 1$

$\underline{i} \cdot \underline{j} = |\underline{i}|\ |\underline{j}| \cos \frac{\pi}{2} = 1 \times 1 \times 0 = 0$

i.e: $\underline{i} \cdot \underline{i} = \underline{j} \cdot \underline{j} = 1$ and $\underline{i} \cdot \underline{j} = \underline{j} \cdot \underline{i} = 0$

4. $\underline{a}$, $\underline{b}$, $\underline{c}$ are vectors.

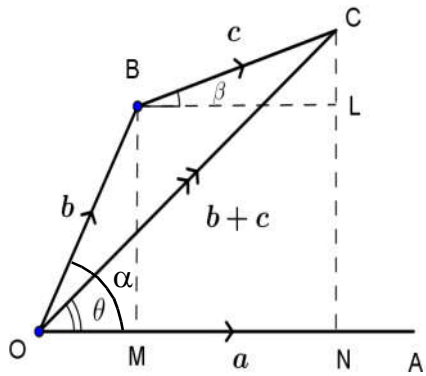
$\underline{a} \cdot (\underline{b} + \underline{c}) = \underline{a}.\underline{b} + \underline{a}.\underline{c}$ (Distributive Law)

Let the angle between

$\underline{a}$ and $\underline{b}$ be α

$\underline{a}$ and $\underline{c}$ be β

$\underline{a}$ and $(\underline{b} + \underline{c})$ be θ



$$\underline{a} \cdot (\underline{b} + \underline{c}) = |\underline{a}| \cdot |\underline{b} + \underline{c}| \cos\theta$$

$$= (OA)\ (OC) \cos\theta$$

$$= (OA) \cdot (ON)$$

$$= (OA)\ (OM + MN)$$

$$= OA.OM + OA \cdot MN$$

$$= OA.\ OB \cos\alpha + OA.\ BC\cos\beta \qquad (MN = BL)$$

$$= \underline{a} \cdot \underline{b} + \underline{a} \cdot \underline{c}$$

$$= \overrightarrow{OA} \cdot \overrightarrow{OB} + \overrightarrow{OA} \cdot \overrightarrow{BC}$$

Therefore $\underline{a} \cdot (\underline{b} + \underline{c}) = \underline{a} \cdot \underline{b} + \underline{a} \cdot \underline{c}$

5. Let $\underline{a} = a_1 \underline{i} + a_2 \underline{j}$ and $\underline{b} = b_1 \underline{i} + b_2 \underline{j}$

$$\underline{a} \cdot \underline{b} = (a_1 \underline{i} + a_2 \underline{j}) \cdot (b_1 \underline{i} + b_2 \underline{j})$$

$$= a_1 \underline{i} \cdot (b_1 \underline{i} + b_2 \underline{j}) + a_2 \underline{j} \cdot (b_1 \underline{i} + b_2 \underline{j})$$

$$= a_1 \underline{i} \cdot b_1 \underline{i} + a_1 \underline{i} \cdot b_2 \underline{j} + a_2 \underline{j} \cdot b_1 \underline{i} + a_2 \underline{i} \cdot b_2 \underline{j}$$

$$= a_1 b_1 + a_2 b_2 \text{ (since } \underline{i} \cdot \underline{i} = \underline{j} \cdot \underline{j} = 1 \text{ and } \underline{i} \cdot \underline{j} = \underline{j} \cdot \underline{i} = 0)$$

Example 7

$\underline{a} = 2\underline{i} - 3\underline{j}$ and $\underline{b} = \underline{i} - 3\underline{j}$ Find the angle between $\underline{a}$ and $\underline{b}$

$\underline{a} \cdot \underline{b} = |\underline{a}|\ |\underline{b}| \cos\theta$

$|\underline{a}| = \sqrt{2^2+3^2} = \sqrt{13}$ $\quad |\underline{b}| = \sqrt{1^2+3^2} = \sqrt{10}$

$\underline{a} \cdot \underline{b} = \sqrt{13} \times \sqrt{10} \cos\theta$ ①

$\underline{a} \cdot \underline{b} = (2\underline{i} - 3\underline{j}) \cdot (\underline{i} - 3\underline{j})$

$= 2\underline{i} \cdot (\underline{i} - 3\underline{j}) - 3\underline{j} \cdot (\underline{i} - 3\underline{j})$

$= 2 + 0 - 0 + 9 = 11$ ②

From (1) and (2) $\sqrt{130} \cos\theta = 11$

$$\cos\theta = \frac{11}{\sqrt{130}}, \theta = \cos^{-1}\left(\frac{11}{\sqrt{130}}\right)$$

Example 8

i. If $\underline{a}$, $\underline{b}$ are two vectors such that $|\underline{a}| = |\underline{b}| = |\underline{a}+\underline{b}|$,

Find the angle between $\underline{a}$ and $\underline{b}$.

ii. Two vectors $\underline{a}$ and $\underline{b}$ are such that $\underline{a}$ is perpendicular to $\underline{a}+\underline{b}$. If $|\underline{b}| = \sqrt{2}|\underline{a}|$,

Show that $(2\underline{a}+\underline{b})$ is perpendicular to $\underline{b}$.

i. $|\underline{a}| = |\underline{b}| = |\underline{a}+\underline{b}|$

$|\underline{a}|^2 = |\underline{a}+\underline{b}|^2$

$\underline{a} \cdot \underline{a} = (\underline{a}+\underline{b}) \cdot (\underline{a}+\underline{b})$ (by definition)

$|\underline{a}|^2 = |\underline{a}|^2 + |\underline{b}|^2 + 2\underline{a}.\underline{b}$

$-|\underline{b}|^2 = 2|\underline{a}|\ |\underline{b}| \cos\theta$

$-|\underline{b}|^2 = 2|\underline{b}|\ |\underline{b}| \cos\theta$

$\cos\theta = -\frac{1}{2}$ $\qquad \theta = \frac{2\pi}{3}$

ii. $(\underline{a}+\underline{b}) \cdot \underline{a} = 0 \Rightarrow \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} = 0$

$|\underline{a}|^2 + \underline{a} \cdot \underline{b} = 0$ ..... ①

$(2\underline{a}+\underline{b}) \cdot \underline{b} = 2\underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{b}$

$= 2\underline{a} \cdot \underline{b} + |\underline{b}|^2$

$= -2|\underline{a}|^2 + |\underline{b}|^2$ (from ①)

$= -2|\underline{a}|^2 + 2|\underline{a}|^2$ (since $|\underline{b}| = \sqrt{2}\underline{a}$)

$= 0$

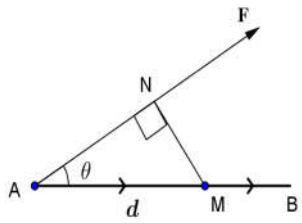
Hence $(2\underline{a}+\underline{b})$ is perpenticular to $\underline{b}$.

Example 9

If a constant force $\underline{\boldsymbol{F}}$ acting on a body moves it a distance $\boldsymbol{d}$ in the direction AB, where $\overrightarrow{AB}$ makes an angle θ with $\underline{\boldsymbol{F}}$, the work done by the force $\underline{\boldsymbol{F}}$ is $\underline{\boldsymbol{F}} \cdot \underline{\boldsymbol{d}}$.

If the point of application of a force $\underline{\boldsymbol{F}} = 2\underline{\boldsymbol{i}} + 3\underline{\boldsymbol{j}}$ makes a displacement $\underline{\boldsymbol{S}} = 5\underline{\boldsymbol{i}} - 3\underline{\boldsymbol{j}}$,

Find the workdone by the force $\underline{\boldsymbol{F}}$.

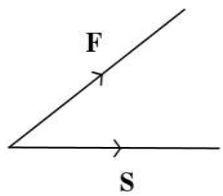


Workdone by $\underline{\boldsymbol{F}}$ is

$= \quad |\underline{\boldsymbol{F}}| .AN$

$= \quad |\underline{\boldsymbol{F}}| .AM \cos\theta \qquad (\overrightarrow{AM} = d)$

$= \quad \underline{\boldsymbol{F}} \cdot \underline{\boldsymbol{d}}$



Workdone by $\underline{\boldsymbol{F}} \cdot \underline{\boldsymbol{S}}$

$= \quad (2\underline{\boldsymbol{i}} + 3\underline{\boldsymbol{j}}) \cdot (5\underline{\boldsymbol{i}} - 3\underline{\boldsymbol{j}})$

$= \quad 2 \times 5 - 3 \times 3 = 10 - 9 = 1$ Joule

1.21 Exercises

1. If $\underline{a} = 3\underline{i} + \underline{j}$ and $\underline{b} = -\underline{i} + 2\underline{j}$, Find the angle between $\underline{\boldsymbol{a}}$ and $\underline{\boldsymbol{b}}$.

2. If $\underline{a} = p\underline{i} + 3\underline{j}$ and $\underline{b} = 2\underline{i} + 6\underline{j}$ are two perpendcular vectors,

   i. find the value of p

   ii. find $|\underline{\boldsymbol{a}}|$ and $|3\underline{\boldsymbol{b}} - \underline{\boldsymbol{a}}|$

   iii. find $\underline{\boldsymbol{a}} \cdot (3\underline{\boldsymbol{b}} - \underline{\boldsymbol{a}})$

   iv. find the angle between $\underline{\boldsymbol{a}}$ and $(3\underline{\boldsymbol{b}} - \underline{\boldsymbol{a}})$

3. Two vectors $\underline{\boldsymbol{a}}$ and $\underline{\boldsymbol{b}}$ are such that $|\underline{\boldsymbol{a}}| = |\underline{\boldsymbol{b}}| = |\underline{\boldsymbol{a}} - \underline{\boldsymbol{b}}|$.

   Find the angle between $\underline{\boldsymbol{a}}$ and $\underline{\boldsymbol{b}}$

4. If $|\underline{\boldsymbol{a}}| = 3$, $|\underline{\boldsymbol{b}}| = 2$ and $|\underline{\boldsymbol{a}} - \underline{\boldsymbol{b}}| = 4$,

   Find (i) $\underline{\boldsymbol{a}} \cdot \underline{\boldsymbol{b}}$

   (ii) $|\underline{\boldsymbol{a}} + \underline{\boldsymbol{b}}|$

5. If $\underline{\boldsymbol{a}}$ and $(\underline{\boldsymbol{a}} + \underline{\boldsymbol{b}})$ are perpendicular vectors to each other,

   Show that $|\underline{\boldsymbol{a}} + \underline{\boldsymbol{b}}|^2 = |\underline{\boldsymbol{b}}|^2 - |\underline{\boldsymbol{a}}|^2$.

6. Using the dot product, show that the diagonals of a rhombus are perpendicular to each other.

7. Show that if $|\underline{\boldsymbol{a}} + \underline{\boldsymbol{b}}| = |\underline{\boldsymbol{a}} - \underline{\boldsymbol{b}}|$ then $\underline{\boldsymbol{a}} \cdot \underline{\boldsymbol{b}} = 0$. Hence, show that if the diagonals of a parallelogram are equal, then it is a rectangle.

8. $\underline{a} = \underline{i} + \sqrt{3}\underline{j}$ where $\underline{\boldsymbol{i}}$ and $\underline{\boldsymbol{j}}$ have the usual meaning. $\underline{\boldsymbol{b}}$ is a vector with magnitude $\sqrt{3}$.

   If the angle between the vectors $\underline{\boldsymbol{a}}$ and $\underline{\boldsymbol{b}}$ is $\frac{\pi}{3}$, Find $\underline{\boldsymbol{b}}$ in he form $\boldsymbol{x}\underline{\boldsymbol{i}} + \boldsymbol{y}\underline{\boldsymbol{j}}$ where $\boldsymbol{x}$ (<0) and $\boldsymbol{y}$ are constants to be determined.

9. AB is a diameter of a circle and P is any point on the circumference of the circle. show that APB is a right angle (use dot product)

10. Using dot product, with the usual notation prove that for any triangle ABC,

    $$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$